Systems of inequalities

1 Order the steps for graphing the inequality $y + 7x \geq 5$.

A Replace the inequality symbol with '$=$' and graph the linear equation.

B If a true statement results shade the half the plane containing the test point. If not, sgade the other half plane.

C Choose a test point from one of the half-planes and not from the line. Substitute its coordinates into the inequaliy.

2 Determine whether the following inequalities are represented using a solid line or a dashed line:

a $y - 3x > 5$ b $x \geq 4$ c $-2y + x \leq -10$ d $y + 3x > 7$

e $-y - 6x > 0$ f $y + 3x \leq -10$ g $y \geq -2$ h $3y + 3x > 4$

3 Consider the system of inequalities.

$$\begin{cases} 2x - y > -6 \\ 2x + 2y < 0 \end{cases}$$

Determine wherther the following points satisfies the system of inequalities.

a $(2, 6)$ b $(-3, 6)$ c $(1, -3)$ d $(-6, 1)$

4 Define the solution set of a system of inequalities.

5 Consider the system of inequalities.

$$\begin{cases} 2x - y < 4 \\ 6x + 3y > 0 \end{cases}$$

a Determine the coordinates of the x- and y-intercepts of the lines $2x - y = 4$ and $6x + 3y = 0$.

b Sketch the system of equations on the same set of axes.

$$\begin{cases} 2x - y = 4 \\ 6x + 3y = 0 \end{cases}$$

c Determine whether the following points [redacted]ies the system of inequalities.

i $(-4,-3)$ ii $(2,2)$ iii $(5,0)$ iv $(2,-7)$

d Hence, graph the solution set to the system of inequalities.

6 Consider the lines $y=4x-3$ and $y=2x-5$.

a Find the point of intersection.

b Graph the region that satisifies both $y<4x-3$ and $y<2x-5$.

7 Consider the system of inequalities:

$$\begin{cases} 3x+y\geq 4 \\ y-2<7x \end{cases}$$

a Graph the system of inequalities.

b Determine whether each of the following statements is true.

i $(1,9)$ is in the solution set because it is contained in the intersection of the two inequalities

ii $(1,9)$ is not in the solution set because it is outside both of the feasible regions.

iii $(1,9)$ is in the solution set because it lies on the solid line.

iv $(1,9)$ is in the solution set because$3\times 1+9\geq 4$.

v $(1,9)$ is not in the solution set because$9-2\geq 7\times 1$

vi $(1,9)$ is not in the solution set because it lies on the dotted line.

8 Sketch the graph of the following systems of inequalities.

a $\begin{cases} x\leq 5 \\ y<3 \end{cases}$ b $\begin{cases} y\leq 3 \\ y\leq 4x+1 \end{cases}$ c $\begin{cases} y\leq x \\ y>-x \end{cases}$ d $\begin{cases} y\leq 3x-4 \\ y>-4x+1 \end{cases}$

e $\begin{cases} 3x+y>5 \\ 3x+y<7 \end{cases}$

9 Sketch the region described by the following:

a Union of $x+y\geq 5$ and $y\leq -5$.

b Intersection of $x+y\geq -2$ and $y>-2x-2$.

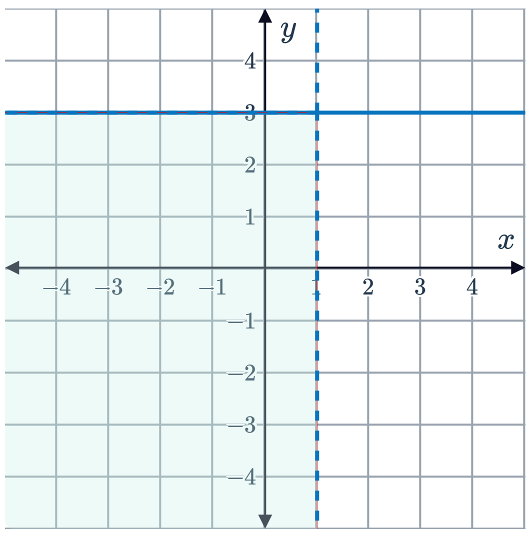
10 Without graphing determine if the following systems of inequalities have no solutions or infinitely many solutions.

a $\begin{cases} 9x - y < 7 \\ 9x - y > 7 \end{cases}$

b $\begin{cases} 2x - y \le 7 \\ 2x - y \ge 7 \end{cases}$

11 Find the system of inequalities that describe each of the following regions:

a



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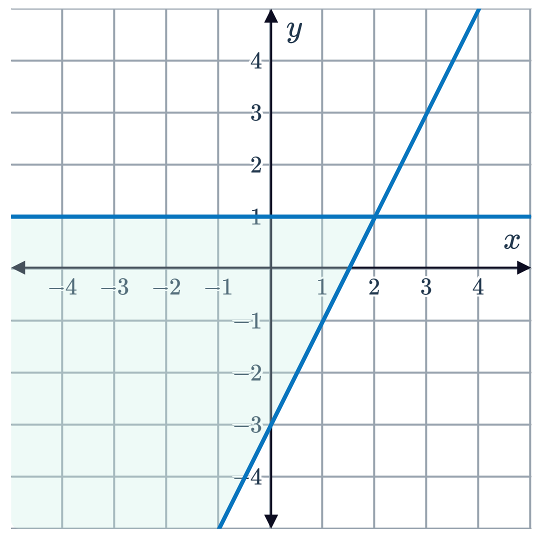

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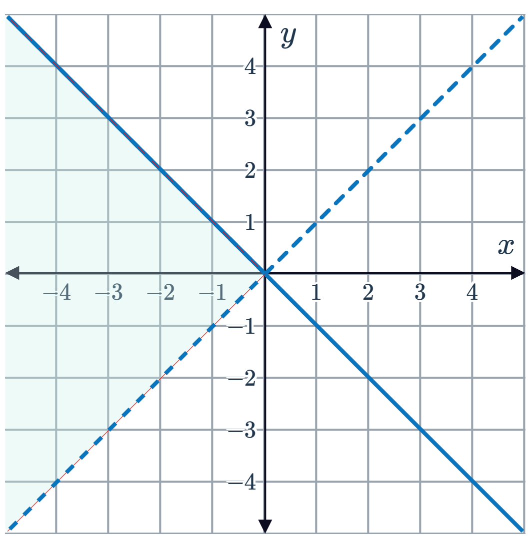

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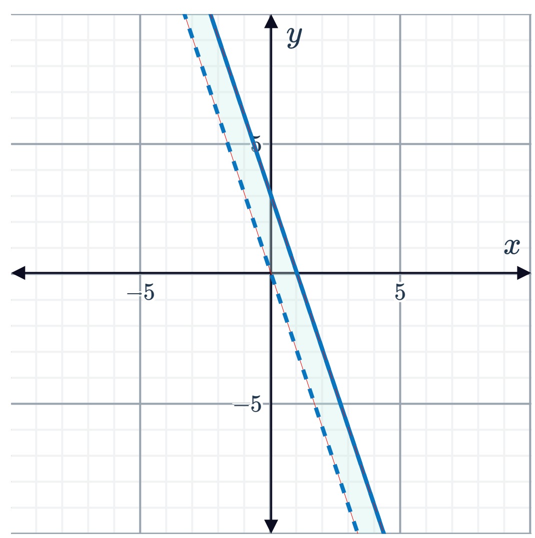

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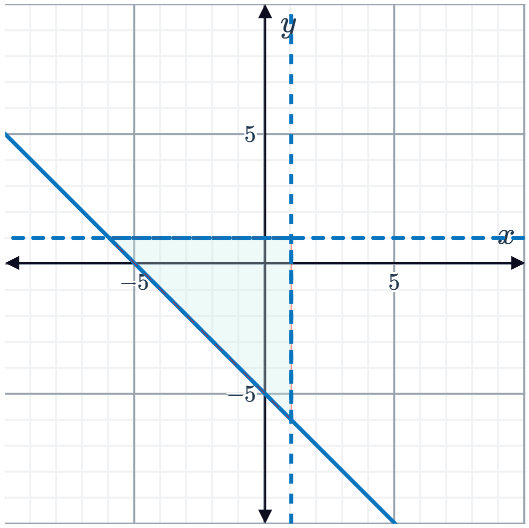
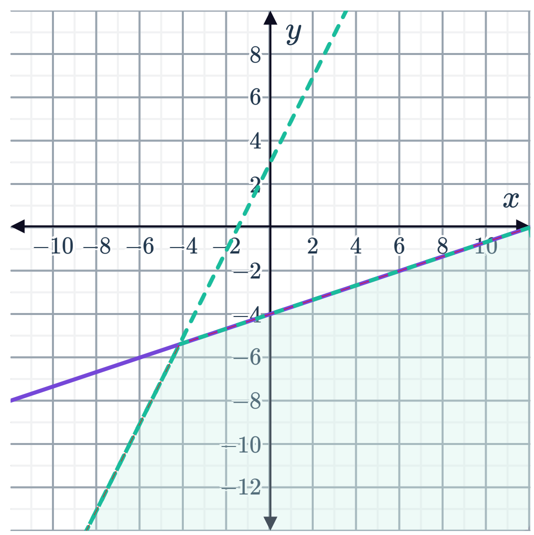


Additional questions

12 Consider the graph of

$$\begin{cases} y < 2x + 3 \\ y \leq \dfrac{1}{3}x - 4 \end{cases}$$



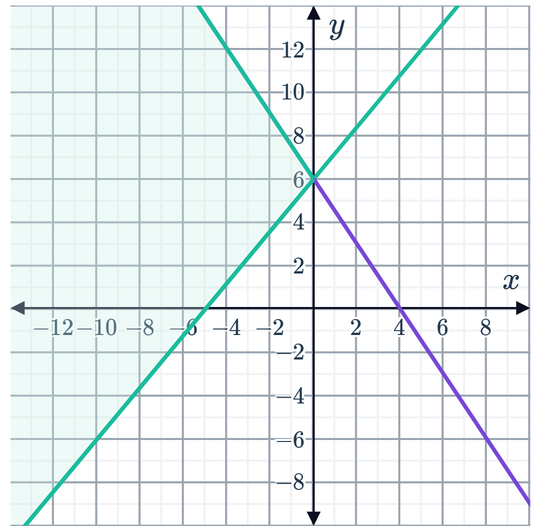
a If the system was changed to

$$\begin{cases} y > 2x + 3 \\ y \leq \dfrac{1}{3}x - 4 \end{cases}$$

sketch a graph of the new system on a coordinate plane.

b Is $(-6, -8)$ a solution to the system in part (a)?

13 Consider the graph of



$$\begin{cases} 5y - 6x \geq 30 \\ 2y + 3x \leq 12 \end{cases}$$

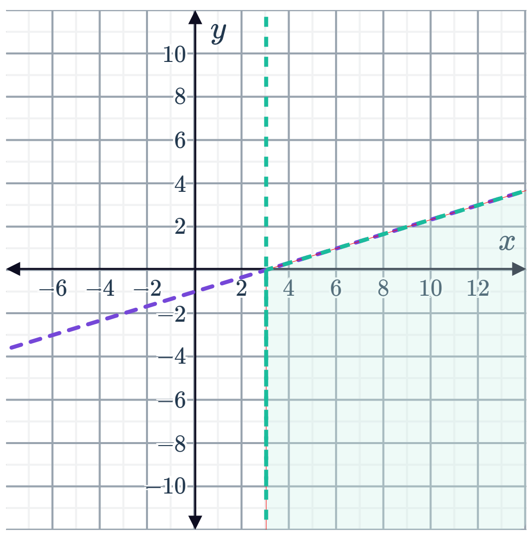
a If the system was changed to

$$\begin{cases} 5y - 6x \geq 30 \\ 2y + 3x < 12 \end{cases}$$

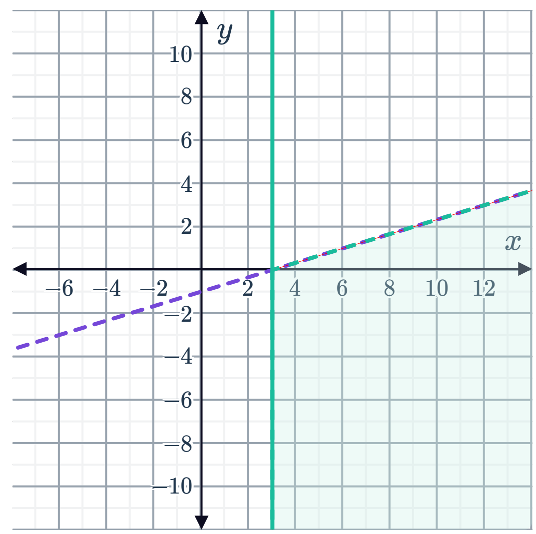
sketch the graph of the new system on the coordinate plane.

b Is $(0, 6)$ a solution to the system in part (a)?

14 **Graph 1**



Graph 2



Graph 1 shows the solution set to the system:

$$\begin{cases} x > 3 \\ y < \dfrac{1}{3}x - 1 \end{cases}$$

a If Graph 2 shows the solution set when one of the inequalities in the system is changed, state the new system of inequalities.

Is $(3, 0)$ a solution to either system? Explain how you know.

15 Yreka Bakery makes two types of cookies: plain and iced. They have enough oven space to bake 15 dozen cookies each day. Each dozen iced cookies requires 0.8 pounds of icing. Yreka Bakery can make no more than 32 pounds of icing per day.
If x is the number of dozens of iced cookies and y is the number of dozens of plain cookies,

16 Throughout university, Jimmy works as a barista, getting paid $\$10$ per hour, and as a mentor getting paid $\$13$ per hour. The number of hours he works in each job can vary from week to week, but he never works more than 27 hours in total each week, and he needs to be able to at least cover his weekly expenses of $\$260$.
If b represents the number of hours worked as a barista and m represents the number of hours worked as a mentor, write a system of inequalities to represent the scenario.

17 David's Pizza makes two types of pizzas: Vegan and Meat Lover's. Based on recent sales they know they will sell at least twice as many Meat Lover's Pizzas as they do Vegan Pizzas. David's Pizza can use 42 pounds of vegan cheese each day, and each Vegan Pizza requires 0.6 pounds of vegan cheese.

a If v is the number of Vegan Pizzas and m is the number of Meat Lover's Pizzas, write a system of inequalities to represent the scenario.

b Is $(28, 70)$ a viable solution in terms of the scenario? Explain.

c Is $(25.25, 71)$ a viable solution in terms of the scenario? Explain.

18 In a team crossfit competition, each team is to be made up of no more than 11 people. For a particular challenge, each team member must complete the course and the team's total time in completing the course must be under 26 minutes. Women take on average 2.6 minutes and men take on average 2.5 minutes to complete the course.

a If w is the number of women and m be the number of men on the team, write a system of inequalities to represent the scenario.

b Sketch a graph of the system of inequalities.

c Give an example of one viable and one non-viable solution to the number of men and women on the team. Explain your answer for both.

19 Throughout university, Tom works as a mentor, getting paid $\$15$ per hour, and as a fitness instructor getting paid $\$18$ per hour. The number of hours he works in each job can vary from week to week, but he never works more than 29 hours in total each week, and he needs to be able to at least cover his weekly expenses of $\$270$.

a If x represents the number of hours Tom works as a mentor and y represents the number of hours he works as a fitness instructor, construct a system of inequalities to represent the scenario.

b Sketch a graph of the system of inequalities.

c If Tom works 6 hours as a mentor in one week, what is the minimum number of hours he can work as a fitness instructor so that he can cover his expenses?

d Is it possible for him to work the same number of hours in both jobs and still be able to cover his expenses?